

Introduction

Cryptocurrency has become one of the fastest-growing financial markets in the world. Due to its highly volatile nature, predicting future price movements and volatility is an important challenge for traders, investors, and financial institutions.

The objective of this project is to analyze historical cryptocurrency market data and develop a machine learning model capable of predicting market volatility. The project uses historical OHLC (Open, High, Low, Close), trading volume, and market capitalization data to understand market behavior.

Before building the prediction model, Exploratory Data Analysis (EDA) was performed to understand the structure of the dataset, identify missing values, detect outliers, analyze statistical properties, and discover relationships among different features.

Objective

The primary objectives of this Exploratory Data Analysis are:

- Understand the dataset structure.
- Analyze numerical and categorical variables.
- Detect missing values and duplicate records.
- Generate descriptive statistics.
- Explore relationships among different features.
- Prepare the dataset for machine learning model development

Dataset Description

Dataset Name:

Cryptocurrency Historical Prices Dataset

Dataset Size:

72,946 Records

Number of Features:

10 Columns

Features Available:

- Open Price
- High Price
- Low Price
- Close Price
- Volume
- Market Capitalization
- Timestamp
- Cryptocurrency Name
- Date

Dataset Information

The dataset contains 72,946 observations and 10 columns. The dataset includes both numerical and categorical variables.

Numerical Features:

- Open
- High
- Low
- Close
- Volume
- Market Capitalization

Categorical Features:

- Timestamp
- Cryptocurrency Name
- Date

The dataset occupies approximately 5.6 MB of memory and is suitable for machine learning applications without requiring large computational resources.

Dataset Structure

Total Records : 72,946

Total Columns : 10

Data Types:

Integer Columns : 1

Float Columns : 6

Object Columns : 3

Memory Usage : Approximately 5.6 MB

Missing Value Analysis

Missing value analysis was performed using the `isnull()` function in Pandas.

Result:

All columns contain zero missing values.

This indicates that the dataset is complete and no missing value treatment was required before model training.

```
[8]
✓ Os # Missing Values
df.isnull().sum()

0
Unnamed: 0  0
open  0
high  0
low  0
close  0
volume  0
marketCap  0
timestamp  0
crypto_name  0
date  0

dtype: int64
```

Statistical Summary

Descriptive statistics were generated using the `describe()` function.

The statistical summary includes:

- Count
- Mean
- Standard Deviation
- Minimum Value
- 25th Percentile
- Median (50%)
- 75th Percentile
- Maximum Value

The statistics show that cryptocurrency prices vary significantly across different currencies. Market capitalization and trading volume also exhibit large variations, indicating the presence of highly active and less active cryptocurrencies.

[7]
✓ 0s

Statistical Summary
df.describe()

...

	Unnamed: 0	open	high	low	close	vo.
count	72946.000000	72946.000000	7.294600e+04	72946.000000	7.294600e+04	7.294600
mean	36472.500000	870.194495	8.964124e+02	844.060640	8.712949e+02	2.207607
std	21057.840705	5231.654470	5.398613e+03	5079.389387	5.235508e+03	9.617885
min	0.000000	0.000000	1.022100e-10	0.000000	8.292000e-11	0.000000
25%	18236.250000	0.167916	1.767999e-01	0.158630	1.682982e-01	8.320618
50%	36472.500000	1.630666	1.717542e+00	1.541486	1.640219e+00	1.098756
75%	54708.750000	26.070557	2.756868e+01	24.791776	2.625195e+01	6.691398
max	72945.000000	67549.735581	1.621883e+05	66458.723733	6.756683e+04	3.509679

Data Quality Assessment

The dataset quality is considered excellent because:

- No missing values were detected.
- Numerical features are properly formatted.
- Categorical features are clearly defined.
- No major inconsistencies were observed during initial exploration.

Therefore, the dataset is suitable for feature engineering and machine learning model development.

Feature Engineering

Feature engineering was performed to improve the predictive capability of the machine learning model.

The following new features were created:

- Price Change = Close Price – Open Price
- Moving Average (MA7) = 7-day moving average of closing price
- Volatility = High Price – Low Price

These engineered features provide additional information regarding short-term price movement and market volatility, helping the model achieve better prediction performance.

Data Visualization

Several visualizations were generated to understand the distribution and relationship among different variables.

The following plots were created:

- Histogram of Closing Price
- Correlation Heatmap
- Feature Importance Graph
- Actual vs Predicted Scatter Plot

These visualizations helped identify important relationships between features and understand the overall behavior of cryptocurrency prices.

Key Observations

The following observations were made during Exploratory Data Analysis:

- The dataset contains 72,946 observations.
- No missing values were found.
- Cryptocurrency prices show significant variation.
- Trading volume and market capitalization differ greatly among cryptocurrencies.
- Engineered features improved the quality of the dataset.
- Random Forest Regressor was selected due to its robustness and ability to model complex nonlinear relationships.

Conclusion

Exploratory Data Analysis provided valuable insights into the cryptocurrency historical price dataset. The data was clean, consistent, and well-structured for machine learning applications.

Feature engineering enhanced the dataset by introducing meaningful variables such as price change, moving average, and volatility. After preprocessing and scaling, the dataset became suitable for training predictive models.

The insights obtained from EDA formed the foundation for developing an accurate cryptocurrency volatility prediction model using Random Forest Regression.

References

- PW Skills Machine Learning Course Material

- Scikit-learn Documentation
<https://scikit-learn.org/>

- Pandas Documentation
<https://pandas.pydata.org/>

- NumPy Documentation
<https://numpy.org/>

- Matplotlib Documentation
<https://matplotlib.org/>